# StampFly v1.1

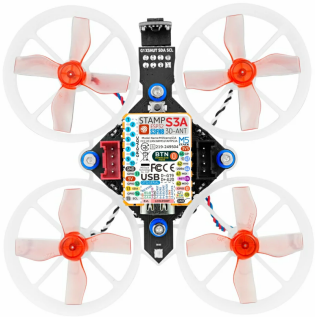
SKU:K138-V11



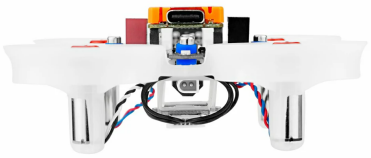


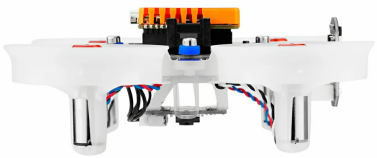




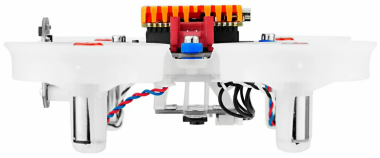



















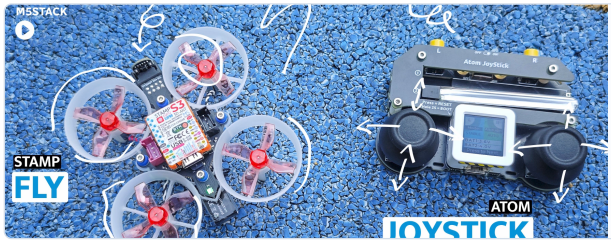

## Description

StampFly v1.1 is a programmable open-source quadcopter kit, an iterative upgrade of the StampFly series. The core improvement is the main control module upgraded from Stamp-S3 to the more powerful Stamp-S3A, with a newly optimized antenna structure design that significantly enhances wireless signal reception performance, resulting in a more stable remote connection.

The drone integrates a BMI270 six-axis gyroscope and a BMM150 three-axis magnetometer for accurate attitude and orientation detection. Paired with a BMP280 pressure sensor and two VL53L3 distance sensors, it achieves stable precise altitude hold and intelligent obstacle avoidance. Onboard buzzer, reset button, and WS2812 RGB LED are used for human-machine interaction and device status indication. It comes with a 320mAh high-voltage battery and four high-speed coreless motors, providing ample power output.

An onboard INA3221AIRGVR chip monitors real-time current and voltage. Two Grove expansion interfaces are reserved for flexible connection of various sensors and peripherals. The product is pre-loaded with a debugging program. When used with the Atom Joystick remote controller, it establishes wireless communication via the ESP-NOW protocol, supporting both manual and automatic flight modes, enabling features like fixed-point hovering and aerial flips. The complete firmware source code is open-source, facilitating secondary development and feature customization. It is suitable for STEM education, scientific research experiments, drone project development, and embedded learning training.

## Tutorial




### StampFly & Atom Joystick Tutorial

This tutorial will introduce you to using M5Burner to flash the factory firmware on StampFly & Atom Joystick, pairing, and the basic operation and indicators of the quadcopter.

# Note

**Charging Notice**

Connect the battery to the charging slot of the Atom Joystick, then connect the data cable to the Atom Joystick to start charging.

**Battery Maintenance**

1. Do not discharge the battery below 3V per cell under load.
2. Do not store a fully charged battery for more than 3 days. For long-term storage, keep the voltage between 3.8V and 3.9V.

# Features

- Stamp-S3A main controller
- BMP280 atmospheric pressure detection
- VL53L3 distance sensor for altitude hold and obstacle avoidance
- 6-axis attitude sensor
- 3-axis magnetometer for direction detection
- 320mAh high-voltage battery
- Current and voltage detection
- Grove expansion interface

# Includes

- 1x Stamp Fly v1.1
- 1x 320mAh battery
- 1x Propeller remover
- 2x Propeller (0.8mm hole diameter)

# Applications

- STEM education
- Open-source DIY creative projects
- Drone R&D
- Intelligent flight algorithm validation

# Specifications

| Specification | Parameter |
| --- | --- |
| Module | Stamp-S3A |
| SoC | ESP32-S3 @ Xtensa® 32-bit LX7 dual-core processor, clock frequency 240MHz |
| USB | USB OTG, USB Serial/JTAG |
| Flash | 8MB |
| Wi-Fi | 2.4 GHz Wi-Fi |
| Drive Motors | 716-17600kv |
| Distance Sensor | VL53L3C (7-bit accuracy, max range 3m) |
| Pressure Sensor | BMP280 (sensing range: 300~1100hPa) |
| 3-Axis Magnetometer | BMM150 |
| 6-Axis Attitude Sensor | BMI270 |
| Interface | I2C / UART |
| Battery | 320mAh battery |
| Battery Output Voltage | 4.35V |
| Flight Time | Approx. 4 minutes |
| Charging Parameters | Input 5V@1A, full charge approx. 55 minutes |
| Monitoring Chip | INA3221AIRGVR voltage/current monitor |
| Buzzer | Onboard 5020 passive buzzer |
| Operating Temperature | 0 ~ 40°C |
| Product Weight | 27.6g |
| Product Size | 73.6 x 73.6 x 32.0mm |
| Package Size | 126.0 x 99.0 x 30.0mm |
| Gross Weight | 80.0g |

# Learn

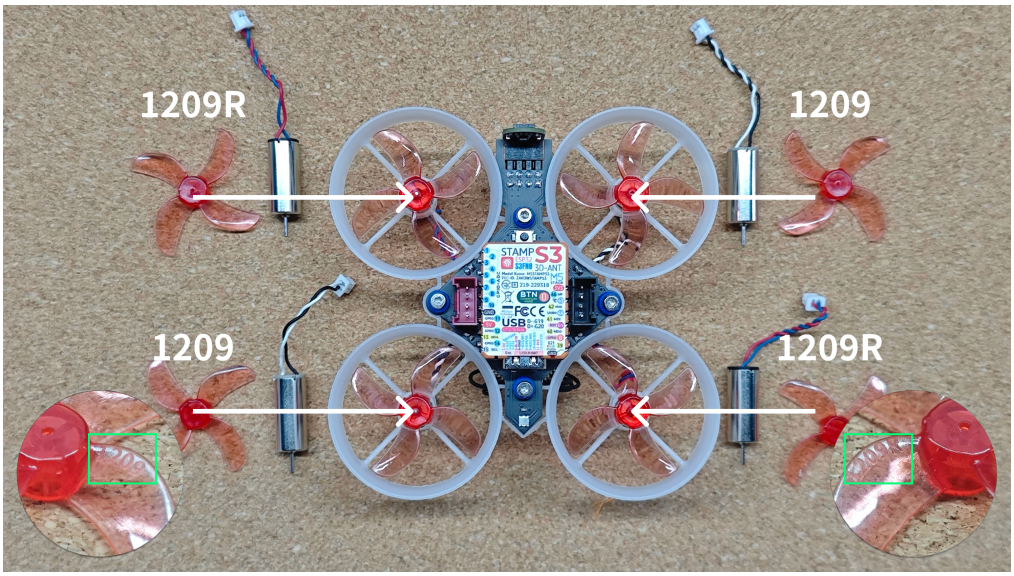
**BMM150 Magnetic Interference**

Products with magnets may interfere with the BMM150 magnetic sensor, causing abnormal readings. When used with M5 main controllers that contain magnets, the magnet must be removed, and the BMM150 sensor should be kept away from strong magnetic fields.

## Propeller Installation Direction Diagram

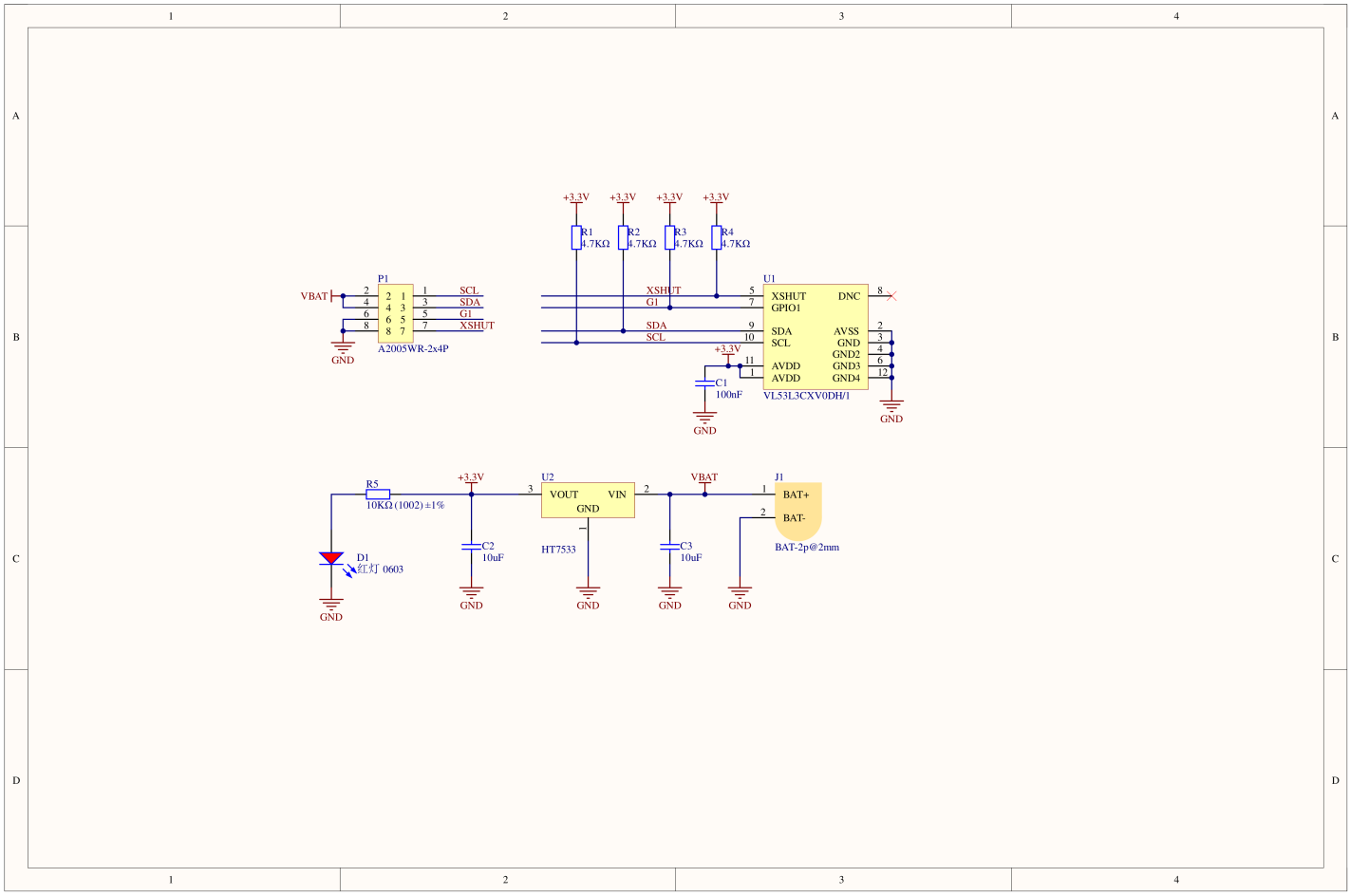
**Note**

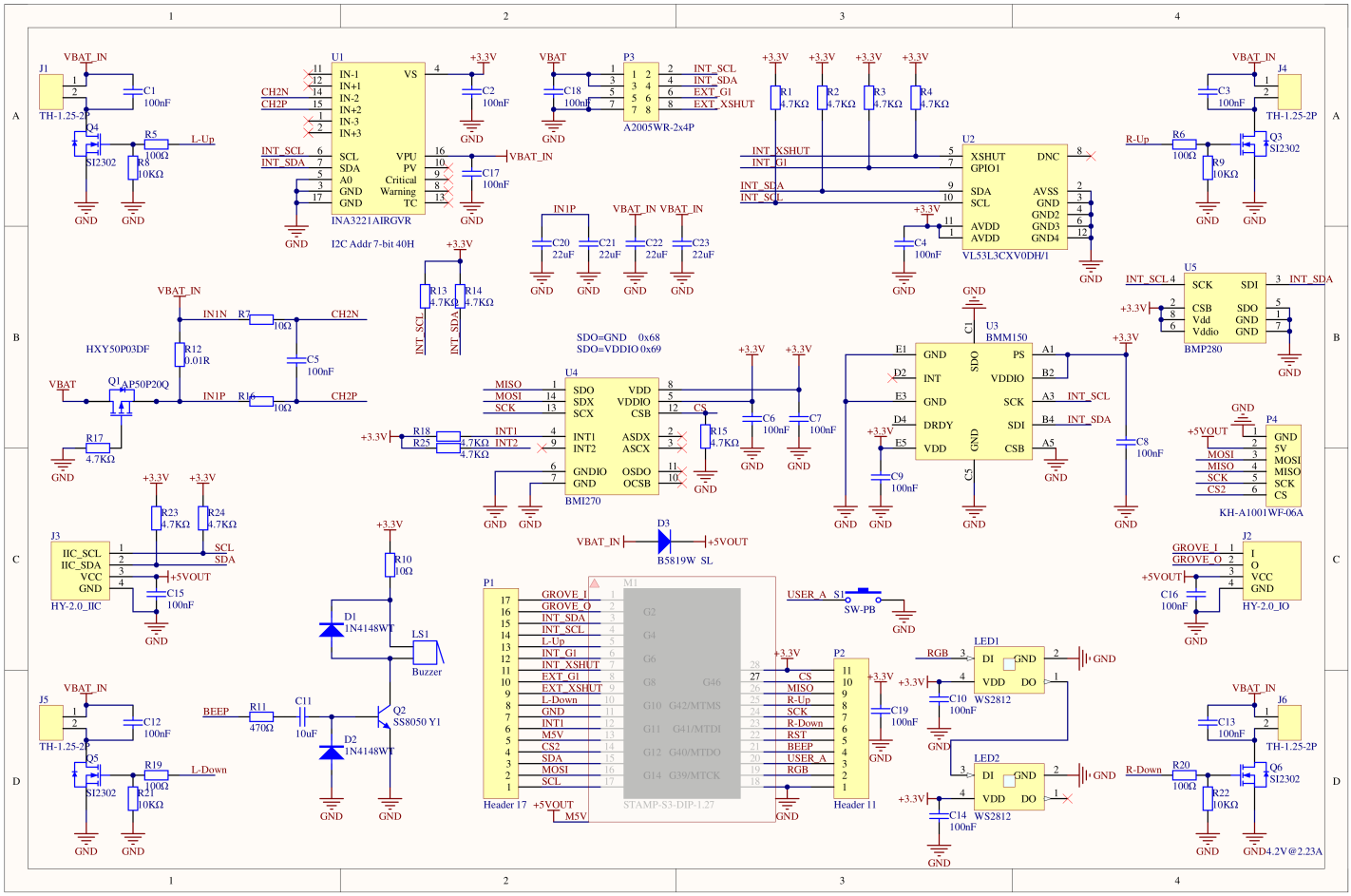
Use the StampFly Repair Kit to replace the propellers. Please check the propeller orientation during installation; incorrect mounting will prevent the device from working properly.

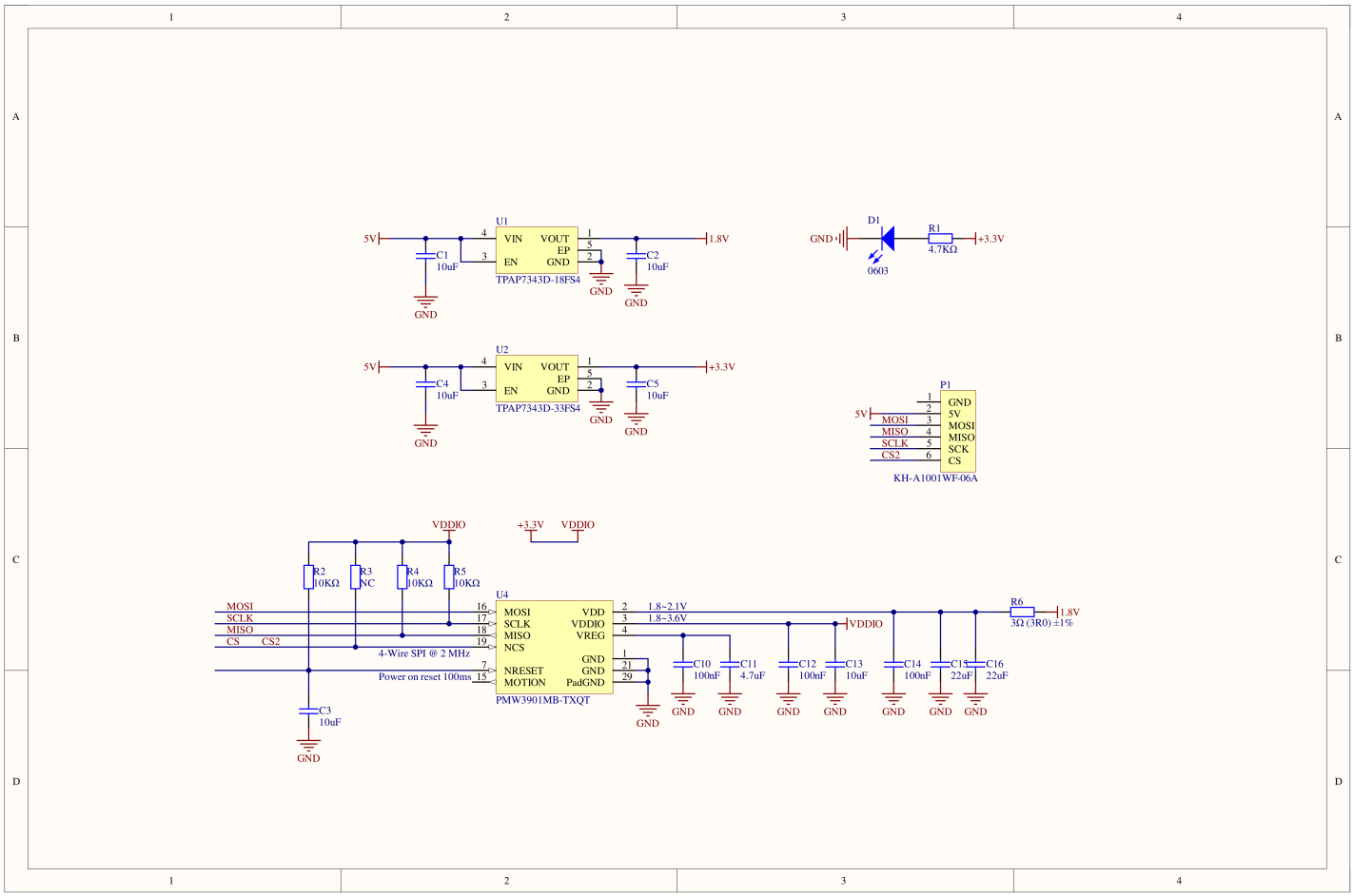



## Schematics

- StampS3_Fly_Hat Schematic PDF
- Stamp_Fly Schematic PDF
- PMW3901MB Schematic PDF




VBAT_IN
J1
TH-1.25-2P
C1
100nF
Q4
SI2302
R5
L-Up
100Ω
R8
10KΩ
GND
U1
IN-1
IN+1
IN-2
IN+2
IN-3
IN+3
VS
VPU
PV
Critical
Warning
TC
SCL
SDA
A0
GND
INA3221AIRGVR
I2C Addr 7-bit 40H
CH2N
CH2P
INT_SCL
INT_SDA
+3.3V
C2
100nF
VBAT_IN
C17
100nF
VBAT
P3
A2005WR-2x4P
C18
100nF
INT_SCL
INT_SDA
EXT_GI
EXT_XSHUT
R1
4.7KΩ
R2
4.7KΩ
R3
4.7KΩ
R4
4.7KΩ
U2
VL53L3CXV0DH/1
XSHUT
GPIO1
DNC
SDA
SCL
AVSS
GND2
GND3
GND4
AVDD
C4
100nF
J4
C3
100nF
R-Up
R6
100Ω
R9
10KΩ
Q3
SI2302
IN1P
C20
22uF
C21
22uF
C22
22uF
C23
22uF
R13
4.7KΩ
R14
4.7KΩ
U5
BMP280
SCK
SDI
CSB
SDO
Vdd
Vddio
HXY50P03DF
IN1N
R7
10Ω
CH2N
R12
0.01R
C5
100nF
Q1
AP50P20Q
R16
CH2P
R17
4.7KΩ
SDO=GND 0x68
SDO=VDDIO 0x69
U4
BMI270
MISO
MOSI
SCK
SDO
SDX
SCX
VDDIO
CSB
INT1
INT2
ASDX
ASCX
GNDIO
OSDO
OCSB
R18
R25
4.7KΩ
R15
4.7KΩ
C6
100nF
C7
100nF
U3
BMM150
E1
D2
E3
D4
E5
A1
B2
A3
B4
A5
INT
VDDIO
SCK
SDI
DRDY
C8
100nF
C9
100nF
P4
KH-A1001WF-06A
+5VOUT
J3
HY-2.0_IIC
IIC_SCL
IIC_SDA
R23
4.7KΩ
R24
4.7KΩ
C15
100nF
R10
10Ω
D1
1N4148WT
LS1
Buzzer
Q2
SS8050 Y1
BEEP
R11
470Ω
C11
10uF
D2
1N4148WT
D3
B5819W SL
P1
Header 17
M1
STAMP-S3-DIP-1.27
G2
G4
G6
G8
G46
G10
G42/MTMS
G11
G41/MTDI
G12
G40/MTDO
G14
G39/MTCK
M5V
USER_A_S1
SW-PB
P2
Header 11
CS
MISO
R-Up
SCK
R-Down
RST
BEEP
USER_A
RGB
LED1
LED2
DI
DO
WS2812
C10
100nF
C14
100nF
C19
100nF
J2
HY-2.0_IO
GROVE_I
GROVE_O
C16
100nF
J5
J6
C12
100nF
C13
100nF
Q5
Q6
R19
L-Down
R20
R21
10KΩ
R22
10KΩ
GND4.2V@2.23A


U1
TPAP7343D-18FS4
VIN
VOUT
EP
EN
GND
5V
1.8V
C1
10uF
C2
10uF
D1
0603
R1
4.7KΩ
+3.3V
U2
TPAP7343D-33FS4
C4
10uF
C5
10uF
P1
KH-A1001WF-06A
MOSI
MISO
SCLK
CS2
CS
5V
U4
PMW3901MB-TXQT
VDDIO
SCLK
NCS
NRESET
MOTION
VREG
PadGND
R2
10KΩ
R3
NC
R4
10KΩ
R5
10KΩ
4-Wire SPI @ 2 MHz
Power on reset 100ms
1.8~2.1V
1.8~3.6V
R6
3Ω (3R0) ±1%
C3
10uF
C10
100nF
C11
4.7uF
C12
100nF
C13
10uF
C14
100nF
C15
22nF
C16
22uF

# PinMap

## I2C Interface

| StampFly v1.1 (Stamp-S3A) | G3 | G4 |
|---|---|---|
| INA3221AIRGVG (0x40) | INA_SDA | INA_SCL |
| BMM150 (0x10) | INA_SDA | INA_SCL |
| BMP280 (0x76) | INA_SDA | INA_SCL |
| VL53L3 (0x29) | INA_SDA | INA_SCL |

## SPI Interface

| StampFly v1.1 (Stamp-S3A) | G14 | G44 | G43 | G46 | G12 |
|---|---|---|---|---|---|
| BMI270 | MOSI | SCK | MISO | CS | |
| PMW3901MB-TXQT | MOSI | SCK | MISO | | CS2 |

## Grove Interface

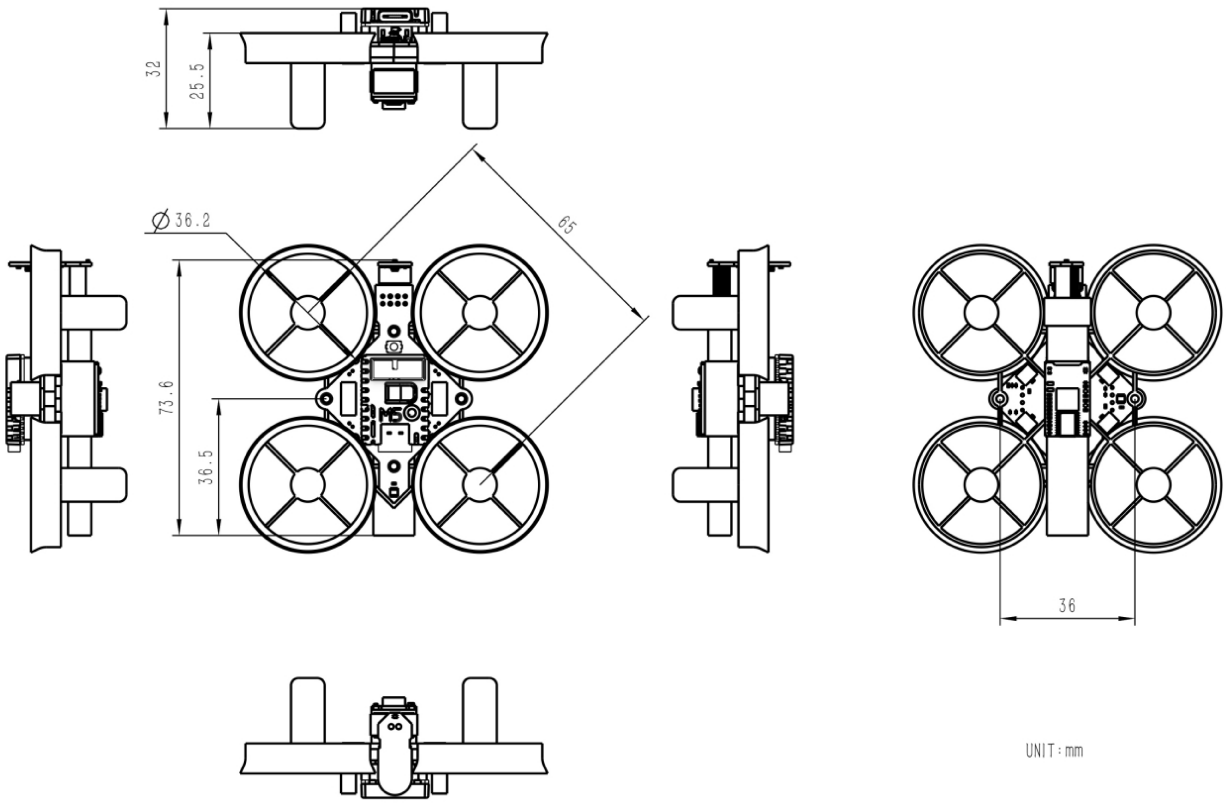
| StampFly v1.1 (Stamp-S3A) | G13 | G15 | G1 | G2 |
|---|---|---|---|---|
| Grove (RED) | SDA | SCL | | |
| Grove (BLACK) | | | GROVE I | GROVE O |

## Buzzer and RGB LED

| StampFly v1.1 (Stamp-S3A) | G40 | G39 |
|---|---|---|
| BEEP | BEEP | |
| WS2812 | | RGB |

# Model Size

- StampFly v1.1 Model Size PDF




## Structure

## Datasheets

- VL53L3(ToF)
- PMW3901MB-TXQT
- BMP280
- BMM150
- BMI270

## Softwares

### Arduino

- StampFly v1.1 Firmware

### Easyloader

| Easyloader | Download Link | Note |
| --- | --- | --- |
| StampFly Firmware Easyloader | download | / |

## Video

- Basic function demonstration of Atom Joystick with Stamp Fly quadcopter

StampFly、AtomJoyStick video.mp4

## Product Comparison

| Product Compare |  StampFly v1.1 |  StampFly |
|---|---|---|
| Core Module | Stamp-S3A | StampS3 |
| Antenna Design | Optimized design, stronger signal | Standard design |
| StampS3 Module Boot Button | Optimized button feel, button spec 4.0 x 3.0 x 2.0mm | Button spec 2.6 x 1.6 x 0.55mm |
| Power Consumption | Optimized for lower power consumption | Standard design |
| Battery Capacity | 320mAh | 300mAh |